Bonus Project 4-2 Ball-on-Beam Project

Implementing PID position control with the classic ball-on-beam setup

Project Goal

Measure the ball's position with an ultrasonic sensor (HC-SR04), and drive a servo (SG90) to keep the ball balanced at a setpoint — e.g. the center of the beam, or 15cm from the sensor.

HC-SR04

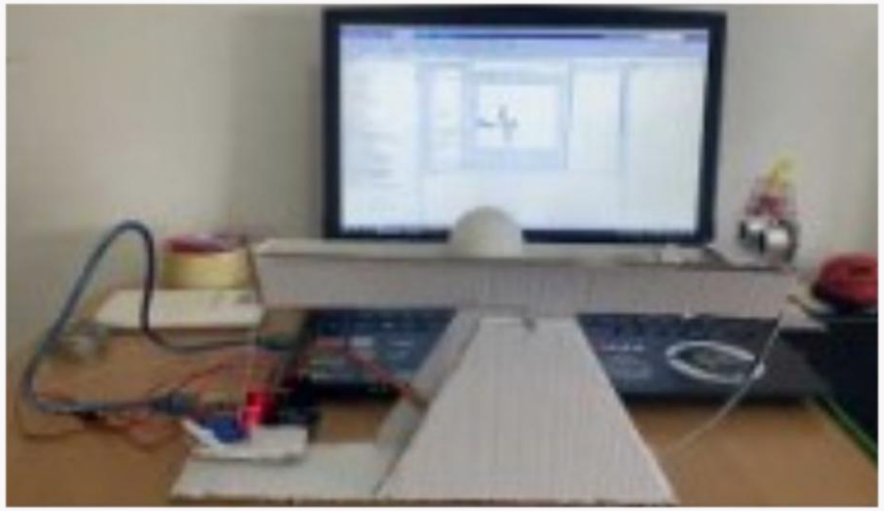
Ultrasonic sensor — measures the ball's distance/position

SG90 Servo

Tilts the beam to push the ball toward the setpoint

Deadline

Bonus Projects 4-2 & 4-3 are due 10/29



Reference video:
https://www.youtube.com/watch?v=tUDcpdTnhn4

Bonus Project 4-2 Implementation Guide

1. Mechanical Assembly

Build the track from cardboard, mount the HC-SR04 firmly at the origin end; connect the SG90 horn to the bottom of the track with steel wire for smooth tilting.

2. Sensor Signal Processing

Read distance data from the HC-SR04. You may need a filter (e.g. Moving Average Filter) to reduce noise.

3. Servo Angle Mapping

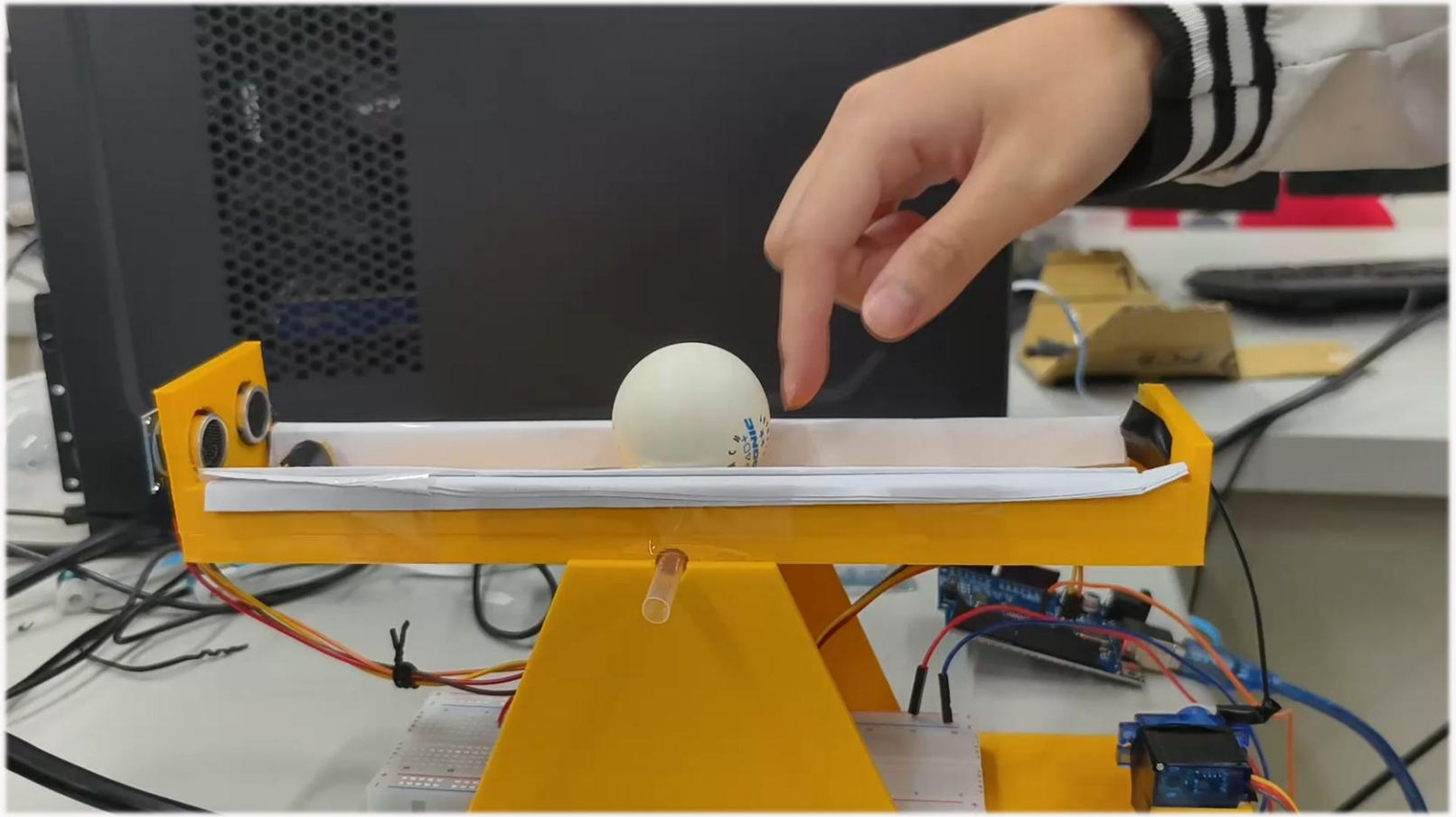
Set a horizontal baseline balance angle (e.g. 90°) and constrain the tilt limits (e.g. 75°–105°) instead of a raw target position.

4. PID Parameter Tuning

Start with Ki = Kd = 0. Increase Kp until the ball oscillates continuously, then tune Kd, and finally Ki.

Materials (from TAs): cardboard, steel/iron wire, a lightweight ping-pong or Styrofoam ball. **Simulation practice:** try tuning Kp/Ki/Kd on TinkerCAD first — the sketch there has the core PID logic you can reference for your physical build.

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Bonus Project 4-3 Free Development Project

Add your own sensors and functionality to the Ball-on-Beam project

Real-time target control

Add a potentiometer to move the ball's target position in real time

Status indicator

An RGB LED or buzzer to show system state: far from target, approaching, or locked on

C# tuning GUI

A C# app to adjust Kp, Ki, Kd from a graphical interface

C# live plotter

A C# app with a real-time graph comparing the setpoint vs. the actual ball position

...and anything else you can dream up!